

Atmospheric Boundary Layer PDE Closure Benchmark Report

Empirical Discovery, Stability Diagnostics, and Galerkin PDE Benchmarks

AtmosphericSlowManifold.jl Pipeline

August 7, 2026

Abstract

This report synthesizes field campaign diagnostics (CASES-99, FLOSS, BLLAST, and SHEBA) and automated equation discovery for atmospheric boundary layer (ABL) turbulence parameterization. We present vertical eddy diffusivity $K_m(z)$ Bayesian credibility ribbons, Richardson number $Ri_g(z, t)$ heatmaps, and Pareto-optimal PDE closure benchmarks solved via spectral Galerkin discretization.

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1 Field Campaign Dataset Diagnostics

Field campaign observations were ingested to derive Obukhov length L , stability parameter $\zeta = z/L$, non-dimensional wind shear $\phi_m(\zeta)$, and gradient Richardson numbers Ri_g .

Table 1: Atmospheric Boundary Layer Field Campaign Dataset Overview

Campaign	Observations	Height Levels	Mean Wind Speed (m s^{-1})	Mean Richardson No.
BLLAST	5600	18	11.335	0.235
SHEBA	2273	0	4.978	0.345
FLOSS	70796	18	28.755	0.212
CASES-99	6538	18	8.41	0.24

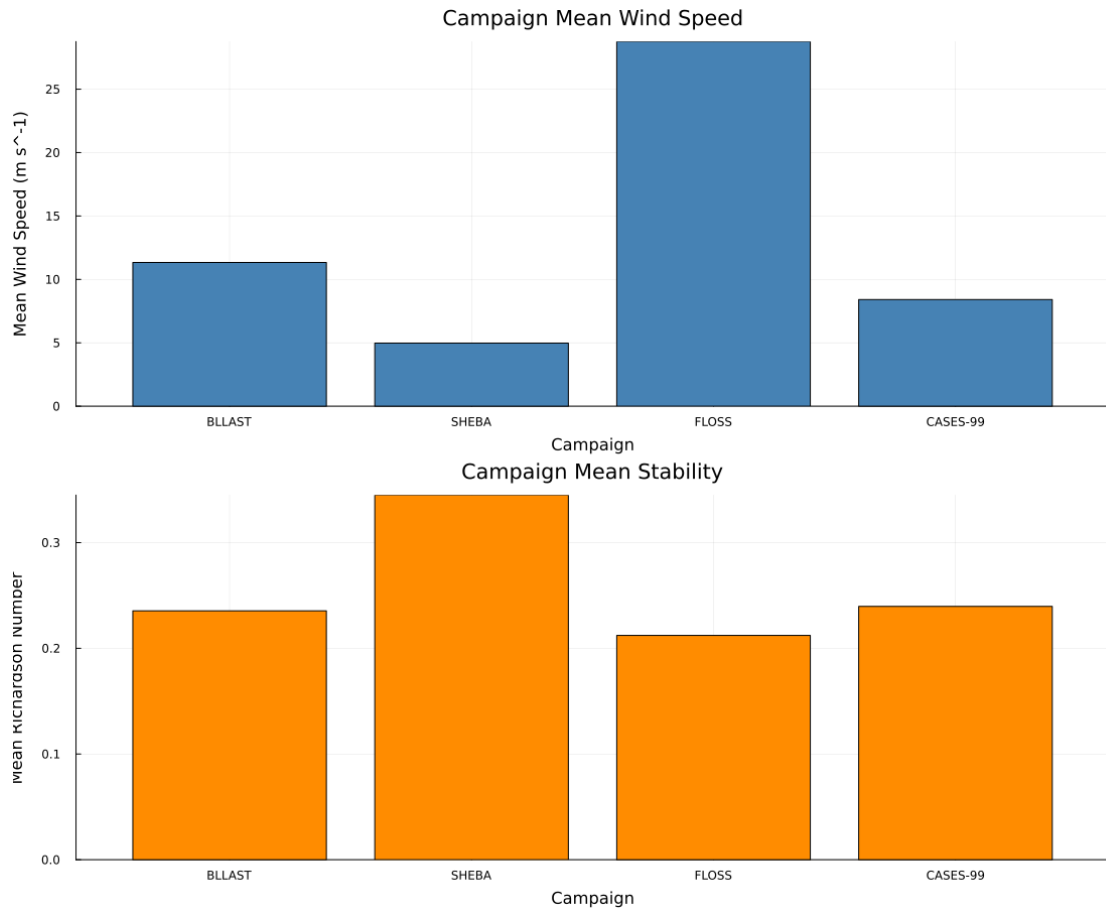


Figure 1: Comparative campaign overview for derived mean wind speed and mean stability metrics.

Table 2: Campaign Production Output Summary

Campaign	Status	Rows	Columns	K _m Ribbon
BLLAST	ok	5600	58	bllast_km_uncertainty.png
SHEBA	ok	2273	27	sheba_km_uncertainty.png
FLOSS	ok	70796	58	floss_km_uncertainty.png
CASES-99	ok	6538	58	cases_99_km_uncertainty.png

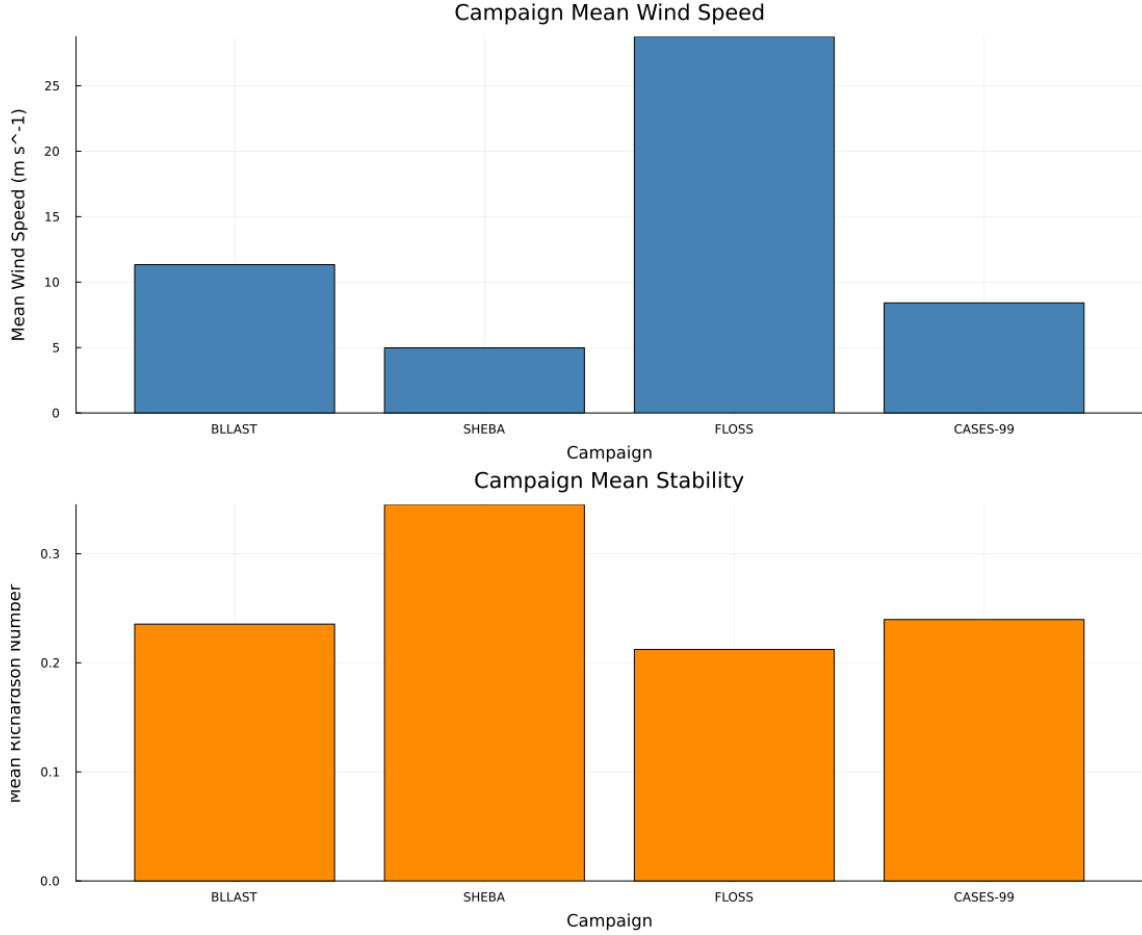
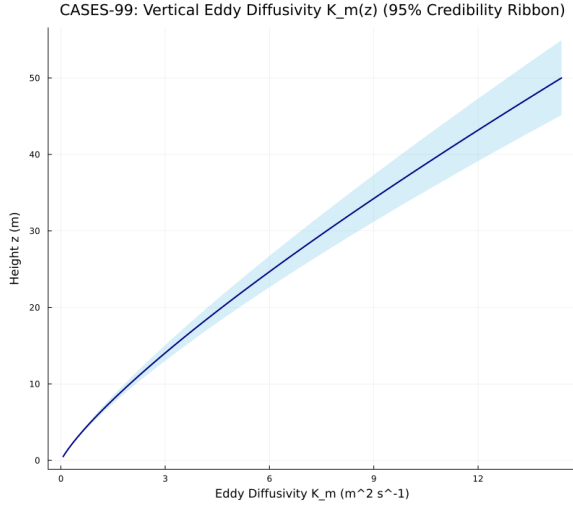


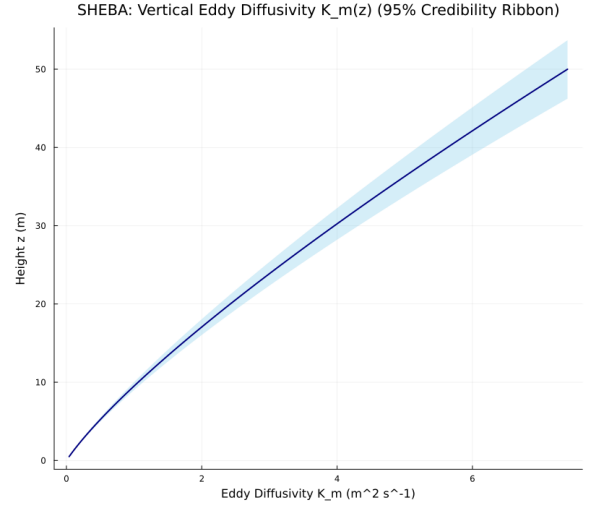
Figure 2: Comparative campaign overview displaying mean wind speed and stability metric profiles across all field sites.

1.1 Vertical Eddy Diffusivity $K_m(z)$ Uncertainty

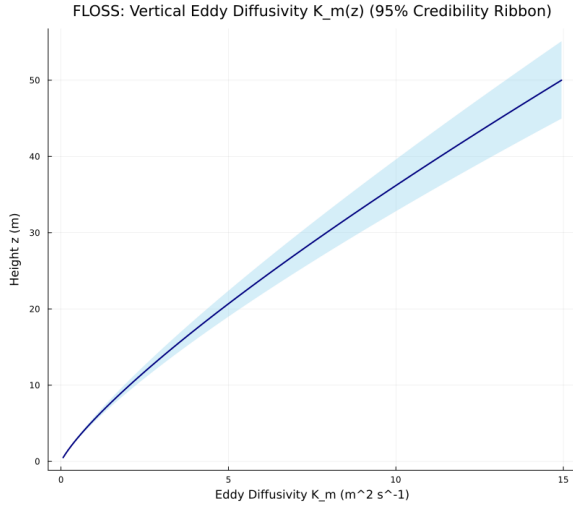
Bayesian parameter sampling propagates uncertainty through Similarity Theory formulations to construct 95% credibility ribbons for $K_m(z)$.



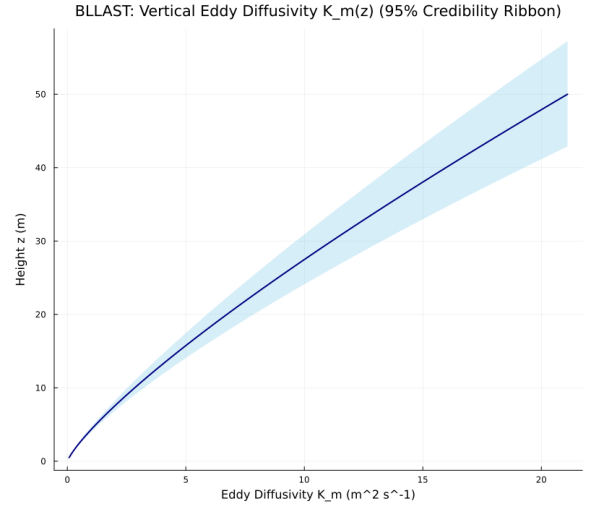
(a) CASES-99 $K_m(z)$ 95% Ribbon



(b) SHEBA $K_m(z)$ 95% Ribbon



(c) FLOSS $K_m(z)$ 95% Ribbon



(d) BLLAST $K_m(z)$ 95% Ribbon

Figure 3: Vertical eddy diffusivity $K_m(z)$ profiles with median predictions and 95% posterior credibility intervals across target field campaigns.

1.2 Stability and Diagnostic Metrics

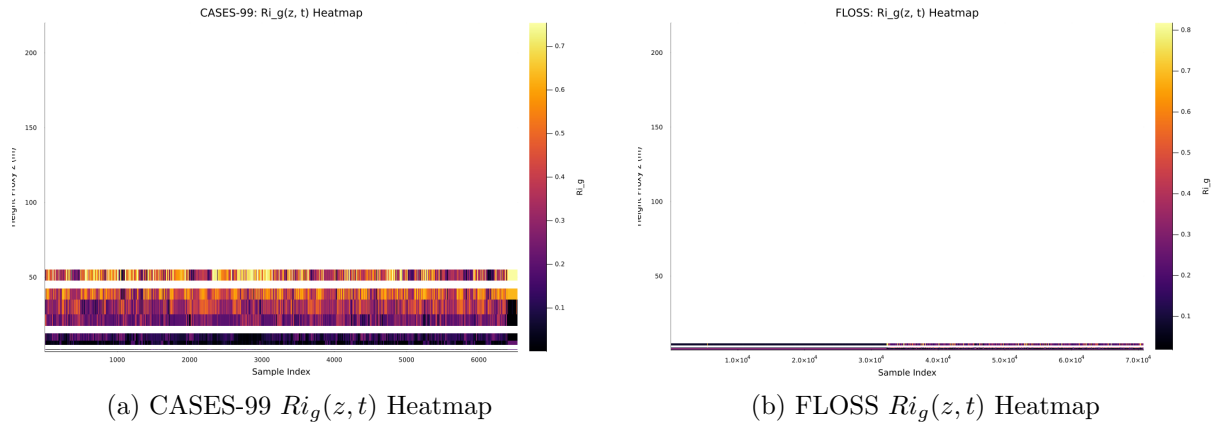


Figure 4: Gradient Richardson number time-height cross sections showing dynamic stability transitions.

2 PDE Closure Benchmark & Model Discovery

Discovered PDE closures are benchmarked against neutral baseline formulations using a 12-mode Gegenbauer Spectral Galerkin discretization integrated with stiff Rosenbrock/BDF ODE solvers.

2.1 Cross-Campaign Discovered Model Summary

Site	Active terms	R^2	Residual norm
cases_99	2	0	8.05183
sheba	2	0	4.76569

2.2 CASES-99 Optimal Pareto Closure

The optimal discovered differential equation governing momentum profile evolution for CASES-99 is:

$$\frac{\partial u}{\partial t} = 0.692696u + 0 \frac{\partial u}{\partial z}$$

Term	Basis	Coefficient
1	u	0.692696
2	$\frac{\partial u}{\partial z}$	0
$R^2 = 0$		
Residual norm=8.05183		

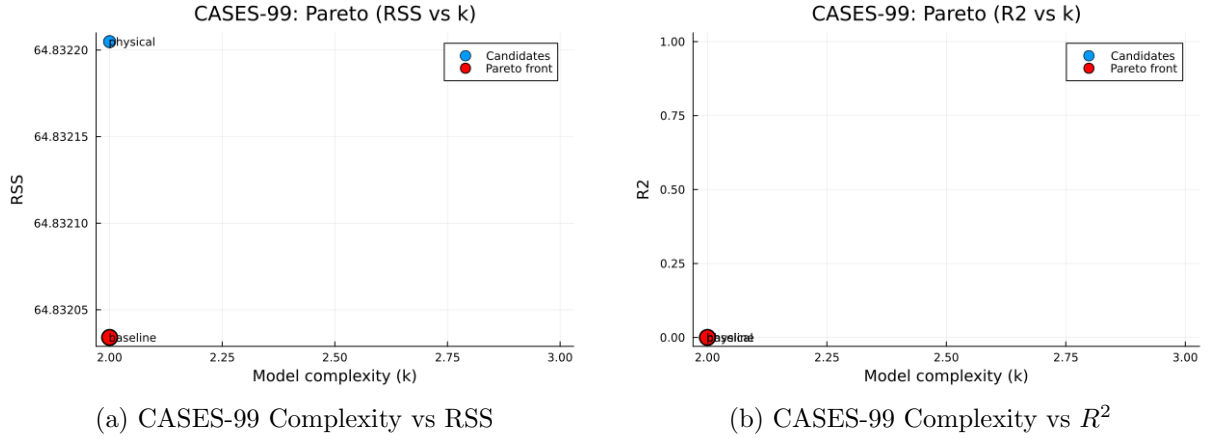


Figure 5: Pareto trade-off fronts between closure sparsity (k) and residual metrics for CASES-99.

2.3 SHEBA Optimal Pareto Closure

The optimal discovered differential equation for SHEBA is:

$$\frac{\partial u}{\partial t} = 0.411338u + 0.0774292 \frac{\partial u}{\partial z}$$

Term	Basis	Coefficient
1	u	0.411338
2	$\frac{\partial u}{\partial z}$	0.0774292
$R^2 = 0$		
Residual norm=4.76569		

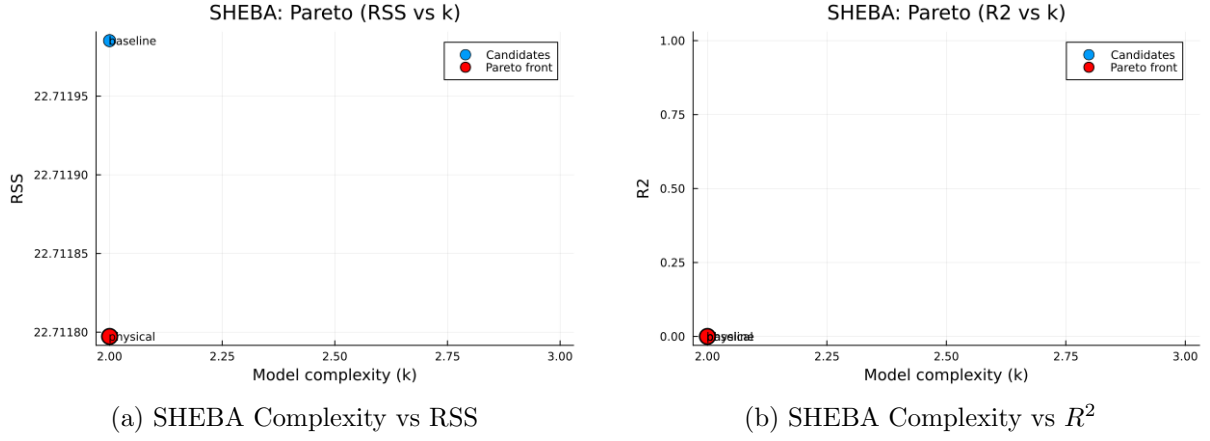


Figure 6: Pareto trade-off fronts between closure sparsity (k) and residual metrics for SHEBA.

2.4 Spectral Galerkin Modal Profiles

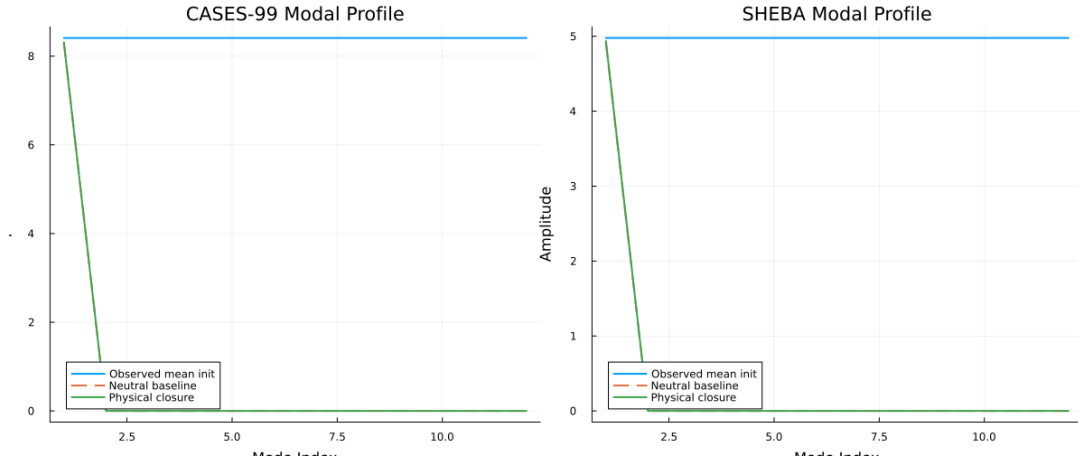


Figure 7: End-state modal profile amplitudes comparing observed mean initial states, neutral baseline predictions, and physical similarity closures.